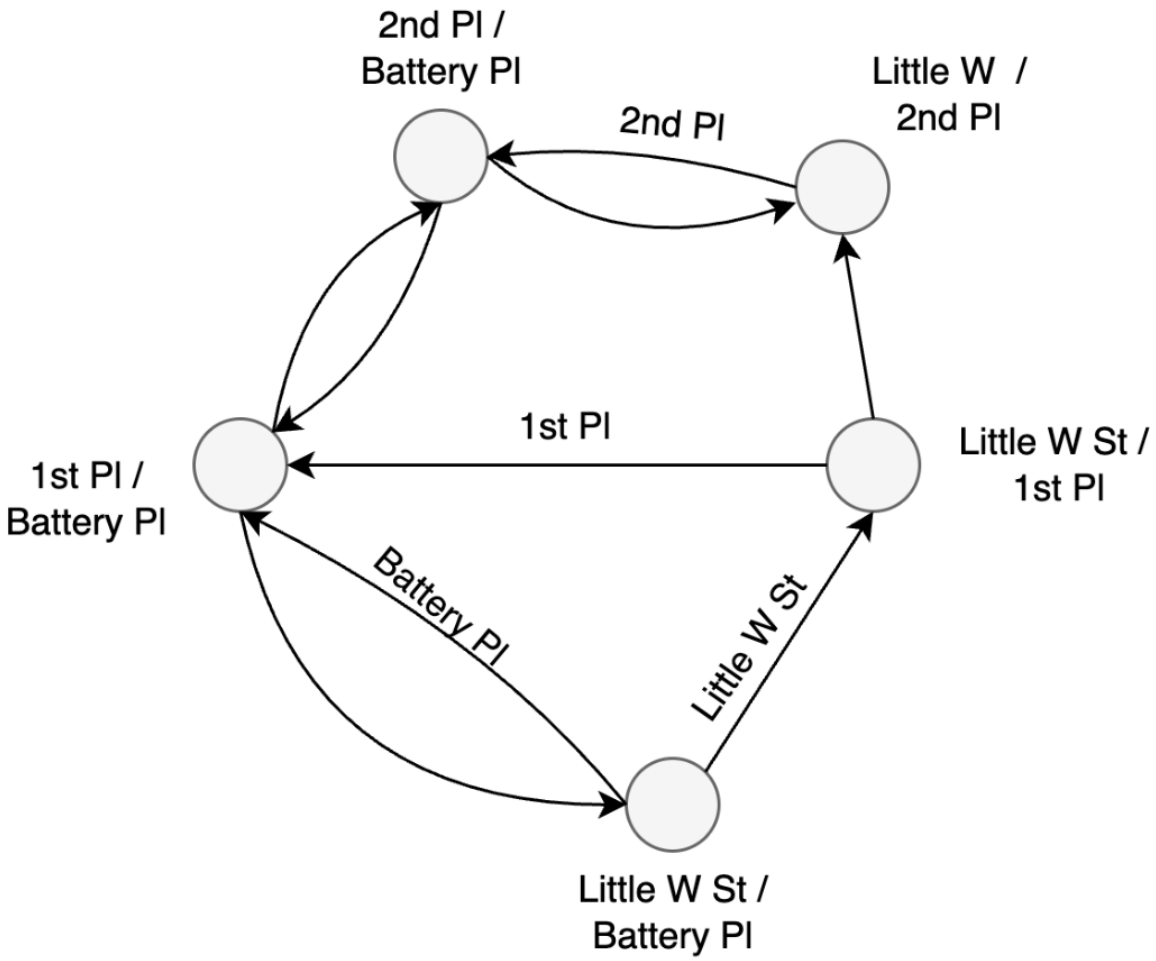


Figure 1.5 A directed graph model of a small subset of lower Manhattan as represented in Google Maps, with vertices representing intersections and edges representing roads with direction of travel.

In the context of digital twins, especially descriptive twins, graph-based models allow us to go beyond what we see and start structuring the world in a way that supports higher-level capabilities like simulation, prediction, and real-time analytics. As you move deeper into digital twin maturity, this structural backbone becomes increasingly important.

1.3.2 Informative digital twin

An *informative digital twin* integrates real-time data streams from the physical system to update the digital representation. This allows users to visualize the current state of the system and make better decisions.